

# Notes on superpatterns: an improved lower bound, short superpatterns for $k = 7, 8$ , and exact values for small alphabets and rosaries

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## Abstract

A permutation  $\sigma \in S_n$  is a  $k$ -superpattern if it contains every  $\pi \in S_k$  as a pattern. Let  $\text{sp}(k)$  be the least such  $n$ . We prove an explicit finite inequality for the number of distinct  $k$ -patterns of any permutation of length  $n$  (Theorem A), and deduce  $\text{sp}(k) \geq (1.0003125 - o(1))k^2/e^2$  with a proof shorter than that of Chroman, Kwan and Singhal (2021), whose constant is 1.000076; a refinement that counts value slots instead of positions (Theorem C) gives  $\text{sp}(k) \geq (1.00483 - o(1))k^2/e^2$ , multiplying the excess over the trivial bound  $k^2/e^2$  by 63. We also give explicit 7- and 8-superpatterns of lengths 23 and 30 (previous best 25 and 33), determine the exact minimal length of  $k$ -superpatterns over the alphabet  $[k+1]$  for  $k \leq 5$  (answering Problem 3 of Hunter for these  $k$ ), and verify Gupta’s rosary conjecture  $r(n) \leq \lfloor n^2/2 \rfloor$  for all  $n \leq 11$ . All computational claims come with certificates that can be re-checked in seconds, and Theorem A, its numerical constant, and the certificates for  $k = 7, 8$  are additionally verified in Lean 4.

## 1 Introduction and summary of results

Let  $\sigma \in S_n$  and  $\pi \in S_k$ . We say  $\sigma$  *contains*  $\pi$  (as a pattern) if there are indices  $t_1 < \dots < t_k$  with  $\sigma(t_i) < \sigma(t_j) \iff \pi(i) < \pi(j)$ . A permutation containing every  $\pi \in S_k$  is a  *$k$ -superpattern* (Arratia [1]; “ $k$ -universal” in [2]). Write  $\text{sp}(k)$  for the minimum length of a  $k$ -superpattern (OEIS A342474) and  $\text{pat}_k(\sigma)$  for the number of distinct  $k$ -patterns contained in  $\sigma$ .

### Previously known.

- $\text{sp}(k) = 1, 3, 5, 9, 13, 17$  for  $k = 1, \dots, 6$  [3].
- Upper bound  $\text{sp}(k) \leq \lceil (k^2 + 1)/2 \rceil$  [3, Thm 6.4], improving  $(k^2 + k)/2$  [4] and  $k^2$  [1].
- Lower bound  $\text{sp}(k) \geq (1.000076/e^2)k^2$  for large  $k$  [2, Thm 1.1], improving the trivial  $\binom{n}{k} \geq k!$  bound  $(1/e^2 - o(1))k^2$  [1] and disproving Arratia’s conjecture that  $k^2/e^2$  is tight.
- For words over the alphabet  $[k+1]$ , Miller’s construction has length  $(k^2 + k)/2$ ; Hunter [5, Thm 1] proved this is asymptotically optimal and asked (Problem 3) whether  $f(k; k+1) = (k^2 + k)/2$  exactly.
- Gupta’s rosary conjecture  $r(n) \leq n^2/2$  is known for even  $n$  [6]; for odd  $n$  only  $r(n) < n^2/2 + n/4 - 1$ .

### Results of this note.

Table 1: Results of this note, with the previously best known value and how each is verified.

| Result                                                                                                                                                                                                                                                              | Previous best                                                                                          | This note                                                                                                         | Verified by                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Lower bound $\text{sp}(k) \geq \lambda k^2/e^2$<br>(Thm A, §2)                                                                                                                                                                                                      | $\lambda = 1.000076$<br>[2]                                                                            | $\lambda = 1.0003125$<br>(excess $\times 4.1$ );<br>shorter proof                                                 | proof; numerics<br>reproduced twice;<br>Theorem A and the<br>constant fully verified in<br>Lean 4 |
| Lower bound, slot refinement<br>(Thm C, §2)                                                                                                                                                                                                                         | $\lambda = 1.000076$<br>[2]                                                                            | $\lambda = 1.00483$<br>(excess $\times 63$ )                                                                      | proof; per-class lemma<br>brute-forced; rate<br>reproduced twice;<br>finite- $k$ certificates     |
| $\text{sp}(7)$ (Thm 2, §3)                                                                                                                                                                                                                                          | $\leq 25$                                                                                              | $\leq 23$                                                                                                         | certificate; brute force;<br>Lean <code>native_decide</code>                                      |
| $\text{sp}(8)$ (Thm 2, §3)                                                                                                                                                                                                                                          | $\leq 33$                                                                                              | $\leq 30$                                                                                                         | certificate; brute force;<br>Lean <code>native_decide</code>                                      |
| Structured 7-superpattern;<br>Arnason's 17 explained (§4)                                                                                                                                                                                                           | 25;<br>unexplained                                                                                     | length 24;<br>tie-broken 3-block<br>word                                                                          | two checkers                                                                                      |
| $f(k; k+1)$ , $k \leq 5$ (Thm 3, §5)                                                                                                                                                                                                                                | $\leq (k^2 + k)/2$                                                                                     | $= (k^2 + k)/2$<br>(Hunter's Problem<br>3)                                                                        | SAT unsatisfiability +<br>witness check                                                           |
| $f(k; k+2)$ , $k = 3, 4$ (§5)<br>Gupta rosaries $r(n)$ (Thm 4, §6)                                                                                                                                                                                                  | —<br>odd $n$ : $< n^2/2 + n/4 - 1$<br>[6]                                                              | $= \text{sp}(k)$<br>$= \lfloor n^2/2 \rfloor$ for<br>$n \leq 5$ ; $\leq \lfloor n^2/2 \rfloor$<br>for $n \leq 11$ | SAT<br>SAT / explicit rosaries +<br>checker                                                       |
| $t(9), t(10)$ (§7)<br>$0.787 \leq c_{21} \leq 1.140$ ; every<br>$\pi \in S_k$ absent from $\Pi_N$ w.h.p. if<br>$k > 2.279\sqrt{N}$ (§7)<br>Every union of $\leq r$ runs<br>contained at $n = O(r^3 \log r k^2)$<br>with failure $e^{-\Omega(n/r^3)}$ (Thm 9,<br>§7) | unknown<br>$0.40 \leq c_{21} \leq e/2$ ;<br>$k > e\sqrt{N}$<br>threads:<br>failure only<br>$e^{-O(k)}$ | $\approx 60, \approx 70-72$<br>threshold of every<br>pattern<br>$\geq 0.1925k^2$<br>speed $N$                     | sampling<br>proof; certificate<br>recomputed                                                      |
| Every block-grid pattern (tilted<br>grids, the family $\mathcal{F}$ , $k = rh$ )<br>contained in a random<br>permutation of length<br>$(\pi/8 + o(1))k^2$ when<br>$\min(r, h) \rightarrow \infty$ (Thm 10, §7)                                                      | $k^2 \text{ polylog } k$<br>(threads;<br>He-Kwan)                                                      | $r$ -independent<br>threshold $0.393k^2$ ,<br>within $\pi/2$ of the<br>conjectured $k^2/4$                        | proof; greedy simulated<br>with copy check                                                        |
| Every $\pi \in S_k$ contained in a<br>random permutation of length<br>$0.757k^2$ w.h.p., failure $e^{-\eta k}$<br>(Thm 11, §7)                                                                                                                                      | $k^2$ (rigid<br>rows =<br>He-Kwan<br>threads);<br>$20k^2$ for<br>quasirandom<br>$\pi$ [7]              | first uniform<br>per-pattern<br>threshold below $k^2$                                                             | reduction proved;<br>constant from<br>quadrature, re-simulated                                    |

| Result                                                                                                                             | Previous best                                             | This note                                                             | Verified by                    |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------|
| Random perm. of length $72k^2$ contains all $\pi$ with $L_\Delta(\pi) \leq k/3 \ln^2 k$ for all but $k/3 \ln k$ shifts (Thm 8, §7) | $20k^2$ for $L_\Delta \leq 3\sqrt{k}$<br>all $\Delta$ [7] | exceptional set $k!e^{-\Theta(k/\ln k)}$ vs $k!e^{-\Theta(\sqrt{k})}$ | proof; overlap lemma simulated |

Result 1 is the main theorem. Results 2–6 are computational; their value is that each is a concrete data point with an easily re-checkable certificate, and together they extend the known tables (A342474 stops at  $k = 6$ ).

## 2 The lower bound

Throughout,  $k$  is odd (the even case follows since  $\text{sp}(k+1) \geq \text{sp}(k)$ ). A  $k$ -subset  $T = \{t_1 < \dots < t_k\} \subseteq [n]$  corresponds bijectively to the *gap vector*  $a = (a_0, \dots, a_k)$ ,  $a_0 = t_1$ ,  $a_j = t_{j+1} - t_j$  ( $1 \leq j < k$ ),  $a_k = n + 1 - t_k$ : a composition of  $n + 1$  into  $k + 1$  positive parts. The *width* of index  $i$  ( $2 \leq i \leq k - 1$ ) is  $b_i = t_{i+1} - t_{i-1} = a_{i-1} + a_i$ .

### 2.1 Stable encodings

The starting point is the encoding idea of [2]: a pattern  $\pi$  occurring at  $T$  can be recorded by giving the positions  $t_j$  for  $j \notin I$  and the *values*  $\pi(i)$  for  $i \in I$ , which is cheaper than giving  $t_i$  whenever the window  $(t_{i-1}, t_{i+1})$  has more than  $k$  integer points. CKS fix a threshold and a set of exactly  $ck$  indices and pay a large-deviation estimate for subsets with few wide windows. The following lemma removes the threshold and the case distinction.

Let  $\mathcal{J}$  be a map  $T \mapsto I(T) \subseteq \{2, \dots, k-1\}$  such that  $I(T)$  contains no two consecutive integers and is *stable*: whenever  $T'$  agrees with  $T$  outside  $I(T)$  (i.e.  $t'_j = t_j$  for all  $j \notin I(T)$ ), we have  $I(T') = I(T)$ .

**Lemma 1.** For every stable  $\mathcal{J}$  and every  $\sigma \in S_n$ ,

$$\text{pat}_k(\sigma) \leq \sum_{T \in \binom{[n]}{k}} \frac{k!}{(k - |I(T)|)!} \prod_{i \in I(T)} \frac{1}{b_i(T) - 1}.$$

*Proof.* For each  $\pi$  contained in  $\sigma$  fix one occurrence  $T = T(\pi)$  and set  $I = I(T)$ ,  $\varphi(\pi) = (I, (t_j)_{j \notin I}, (\pi(i))_{i \in I})$ .  $\varphi$  is injective: from  $\sigma$  and  $(t_j)_{j \notin I}$  we know the relative order of the values  $\sigma(t_j)$ ,  $j \notin I$ ; from  $(\pi(i))_{i \in I}$  we know which  $k - |I|$  ranks in  $[k]$  are taken by the indices  $j \notin I$ ; together these determine  $\pi$ . Group the codes by  $\psi = (I, (t_j)_{j \notin I})$ ; each  $\psi$  admits at most  $k!/(k - |I|)!$  third components. Let  $\text{ext}(\psi)$  be the set of  $T' \in \binom{[n]}{k}$  agreeing with  $\psi$  off  $I$ . Since  $I$  has no two consecutive elements,  $t'_i$  ranges independently over the open interval  $(t_{i-1}, t_{i+1})$  for  $i \in I$ , so  $|\text{ext}(\psi)| = \prod_{i \in I} (b_i - 1)$ , where  $b_i$  is determined by  $\psi$ . By stability, every  $T' \in \text{ext}(\psi)$  has  $I(T') = I$ , so  $T'$  determines  $\psi$  and the sets  $\text{ext}(\psi)$  are pairwise disjoint. Hence

$$\begin{aligned} \sum_{\psi} \frac{k!}{(k - |I|)!} &= \sum_{\psi} \sum_{T' \in \text{ext}(\psi)} \frac{k!}{(k - |I|)! |\text{ext}(\psi)|} \\ &\leq \sum_{T'} \frac{k!}{(k - |I(T')|)!} \prod_{i \in I(T')} \frac{1}{b_i(T') - 1}. \quad \blacksquare \end{aligned}$$

## 2.2 The even rule and the tilt

Take  $I(T) = \{i \text{ even} : b_i - 1 > k\}$ . It is non-adjacent, and stable because it depends only on the odd-indexed positions  $t_{i\pm 1}$ , none of which lie in  $I(T)$ . Using  $k!/(k - |I|)! \leq k^{|I|}$ , Lemma 1 gives

$$\text{pat}_k(\sigma) \leq \sum_T W(T), \quad W(T) = \prod_{i \text{ even}} f(b_i), \quad f(b) = \min\left(1, \frac{k}{b-1}\right). \quad (1)$$

In gap coordinates,  $W(a) = \prod_{j=1}^{(k-1)/2} f(a_{2j-1} + a_{2j})$ : a product over *disjoint* pairs of gaps. For any  $x \in (0, 1)$ , since  $\sum a = n + 1$  on every term,

$$\begin{aligned} \sum_T W(T) &= x^{-(n+1)} \sum_{a: \sum a = n+1} x^{\sum a} W(a) \leq x^{-(n+1)} \sum_{a \in \mathbb{Z}_{\geq 1}^{k+1}} x^{\sum a} W(a) \\ &= x^{-(n+1)} \left(\frac{x}{1-x}\right)^2 \prod_{j=1}^{(k-1)/2} \sum_{a, a' \geq 1} x^{a+a'} f(a+a'). \end{aligned} \quad (2)$$

(The free gaps  $a_0, a_k$  contribute  $\sum_{a \geq 1} x^a = x/(1-x)$  each.)

**Lemma 2.** For  $x \in (0, 1)$  and  $k \geq 1$ ,  $\sum_{a, a' \geq 1} x^{a+a'} f(a+a') = \left(\frac{x}{1-x}\right)^2 (1-x^k)$ .

*Proof.* Writing  $s = a + a'$ , there are  $s - 1$  pairs with sum  $s$ . Split at  $s = k + 1$ :

$$\sum_{s=2}^{k+1} (s-1)x^s + \sum_{s \geq k+2} (s-1)x^s \frac{k}{s-1} = \left[ \sum_{s \geq 2} (s-1)x^s - \sum_{s \geq k+2} (s-1)x^s \right] + \frac{k x^{k+2}}{1-x}.$$

Now  $\sum_{s \geq 2} (s-1)x^s = x^2/(1-x)^2$  and  $\sum_{s \geq k+2} (s-1)x^s = x^{k+2}[(k+1)/(1-x) + x/(1-x)^2]$ . Collecting terms over the common denominator  $(1-x)^2$ :

$$\frac{x^2 - x^{k+2}((k+1)(1-x) + x) + kx^{k+2}(1-x)}{(1-x)^2} = \frac{x^2 - x^{k+2}}{(1-x)^2} = \left(\frac{x}{1-x}\right)^2 (1-x^k).$$

■

Combining (1), (2) and Lemma 2:

**Theorem A.** For every  $\sigma \in S_n$ , every odd  $k \geq 3$  and every  $x \in (0, 1)$ ,

$$\text{pat}_k(\sigma) \leq x^{-(n+1)} \left(\frac{x}{1-x}\right)^{k+1} (1-x^k)^{(k-1)/2}. \quad (\text{A})$$

Note that dropping the factor  $(1-x^k)^{(k-1)/2}$  and optimizing  $x$  recovers exactly the trivial bound  $\binom{n}{k}$  up to lower-order terms; the whole gain is in that factor.

## 2.3 Asymptotics

Put  $n = \lambda k^2/e^2$  and  $x = e^{-\theta/k}$  with  $\theta > 0$  fixed. Then  $-(n+1)\log x = (n+1)\theta/k$ ,  $\log \frac{x}{1-x} = \log \frac{k}{\theta} - \frac{\theta}{2k} + O(k^{-2})$ , and  $\log k! = k \log k - k + O(\log k)$ ; hence

$$\frac{1}{k} \log \frac{\text{RHS(A)}}{k!} = \underbrace{\frac{\lambda \theta}{e^2} - \log \theta + 1 + \frac{1}{2} \log(1 - e^{-\theta})}_{=: g(\theta, \lambda)} + O\left(\frac{\log k}{k}\right). \quad (\text{A}')$$

$g(\cdot, \lambda)$  is strictly convex on  $(0, \infty)$  (it is  $\lambda\theta/e^2 - \log \theta$  plus a convex function), with a unique minimizer  $\theta^*(\lambda)$  solving  $\lambda/e^2 = 1/\theta - \frac{1}{2}e^{-\theta}/(1 - e^{-\theta})$ , and  $\min_{\theta} g(\theta, \lambda)$  is continuous and strictly increasing in  $\lambda$ . Numerically (mpmath, 30 digits, two independent implementations):

$$\begin{aligned}\theta^*(1) &= 7.37192\dots, & g(\theta^*(1), 1) &= -3.1174 \times 10^{-4}, \\ \lambda_A := \text{root of } \min_{\theta} g(\theta, \lambda) &= 1.00031251\dots, & \theta^*(\lambda_A) &= 7.36959\dots\end{aligned}$$

Hence for every fixed  $\lambda < \lambda_A$  and all large  $k$ , every  $\sigma \in S_n$  with  $n \leq \lambda k^2/e^2$  satisfies  $\text{pat}_k(\sigma) \leq e^{-ck}k!$  for some  $c = c(\lambda) > 0$ ; in particular  $\sigma$  is not a  $k$ -superpattern.

**Corollary.**  $\text{sp}(k) \geq (1.0003125 - o(1))k^2/e^2$ .

For a fully explicit finite check, at  $k = 100001$ ,  $\lambda = 1.00031$ ,  $\theta = \theta^*(\lambda)$ , the exact right-hand side of (A) gives  $\frac{1}{k} \log(\text{RHS}/k!) = -1.09 \times 10^{-5} < 0$ .

*Comparison with [2].* Their encoding specifies values for exactly  $ck$  even indices ( $c = 0.00075$ ) whose widths exceed  $dn/k$  ( $d = 8.180$ ), and needs their Lemma 2.1 (a Chernoff bound showing that all but  $O(\lambda^k) \binom{n}{k}$  subsets have that many wide windows); this yields  $\log \lambda_{\text{CKS}} \approx c \log(d/e^2) \approx 7.6 \times 10^{-5}$ . In Theorem A the right-hand side of Lemma 1 is an *expectation of a product*, so after the exponential tilt every wide window contributes its full saving  $\min(1, \theta/B)$ ,  $B \sim \Gamma(2, 1)$  in the limit, with  $\mathbb{E} \min(1, \theta/B) = 1 - e^{-\theta}$ ; no case split is needed. The tilt itself costs only  $\tau - 1 - \log \tau \approx 2.7 \times 10^{-6}$  where  $\tau = \theta\lambda/e^2 \approx 0.9977$ .

We stress that CKS anticipated this: their concluding remarks note that “with a more sophisticated argument one could take both even and odd  $i$  into account” and that “presumably one could prove exact large deviation bounds”, but that fully optimizing “would be very complicated”. Lemma 1 is what makes the optimization short: once the bound is written as a sum of products over subsets, the exponential tilt does the large-deviation work exactly and for free. The novelty is therefore in the formulation, not in a new mechanism; a priority search (citations of [2], [5], [3]; arXiv full text; OEIS A342474; Bóna [8]) found no improvement of the constant 1.000076 in the literature through August 2026.

## 2.4 The value-slot refinement: Theorem C

Lemma 1 charges index  $i \in I$  the factor  $1/(b_i - 1)$  because  $t_i$  has  $b_i - 1$  possible positions, each of which could in principle give a different pattern. But the pattern only depends on where  $\sigma(t_i)$  falls among the other chosen values, so the number of distinct patterns obtainable by moving  $t_i$  is at most the number of *value slots* actually hit by  $\sigma$  on the window. For  $T$  with  $I(T) = \{\iota_1 < \dots < \iota_r\}$  let  $V(T) = \sigma(\{t_j : j \notin I(T)\})$  and let  $M_i(T)$  be the number of distinct slots relative to  $V(T)$  (a slot of  $u$  is  $\#\{v \in V : v < u\}$ ) hit by the values  $\sigma(p)$ ,  $p \in (t_{i-1}, t_{i+1})$ ).

**Lemma 1 .** For every stable rule  $\mathcal{J}$ ,

$$\text{pat}_k(\sigma) \leq \sum_T \min\left(1, \prod_{j=1}^{r(T)} \frac{\min(M_{\iota_j}(T) + j - 1, k)}{b_{\iota_j}(T) - 1}\right).$$

*Proof.* As in Lemma 1, group witnesses into classes  $\text{ext}(\psi)$ ; the number of distinct patterns in a class is trivially at most  $|\text{ext}(\psi)| = \prod (b_i - 1)$ . For the other bound, insert the  $I$ -values  $v_j = \sigma(t'_{\iota_j})$  one at a time into the sorted list of  $V$  and let  $s_j$  be the slot of  $v_j$  relative to  $V_j = V \cup \{v_1, \dots, v_{j-1}\}$ ; the sequence  $(s_1, \dots, s_r)$  determines the pattern. A prefix  $(s_1, \dots, s_{j-1})$  determines the  $V$ -slot  $c_j$  of

each earlier  $v_l$  and their relative order within each  $V$ -slot; for a window value  $u$  in  $V$ -slot  $c$  one has  $\text{slot}(u \mid V_j) = c + \#\{l < j : c_l < c\} + \#\{l < j : c_l = c, v_l < u\}$ , and the last term lies in  $[0, n_c]$  with  $n_c = \#\{l < j : c_l = c\}$ . Hence, over *all* realizations of the prefix,  $s_j$  takes at most  $\sum_{c \text{ hit}} (1 + n_c) \leq M_{\iota_j} + j - 1$  values, and also at most  $|V_j| + 1 \leq k$ . Multiply over  $j$ . ■

(The per-factor cap  $b_{\iota_j} - 1$  would be false here: realizations of the same prefix with different values  $v_l$  send a fixed window position to different slots. We caught this in a first draft by enumerating every class on the  $\text{sp}(6) = 17$  permutation; the corrected bound has no violations on all permutations tested.)

The slot counts are controlled uniformly in  $\sigma$  by a domination argument. Under the geometric tilt,  $T$  is the set of the first  $k$  points of a Bernoulli( $1-x$ ) site process  $B$  on  $[n]$ ; the off- $I$  values are contained in  $\sigma(B \setminus [t_{i-1}, t_{i+1}]) \cup \{\sigma(t_{i-1}), \sigma(t_{i+1})\}$ , and the number of slots hit relative to  $\sigma(B \setminus [t_{i-1}, t_{i+1}])$  equals  $1 + \sum_{\ell} \mathbf{1}[B \cap A_{\ell} \neq \emptyset]$  where the  $A_{\ell}$  (positions of the values lying strictly between consecutive window values) are *disjoint* sets outside the window. These indicators are independent, independent of  $B$  inside the window, and by Jensen have total mean at most  $m(b) = (b-2)(1-x^{(n-b-1)/(b-2)})$ . A Chernoff bound per window and a union bound over the  $O(kn)$  windows then give, for every  $\sigma \in S_n$ ,  $x \in (0, 1)$ , threshold  $\beta$  and  $\varepsilon > 0$ :

**Theorem C.** With  $q = 1-x$ ,  $h(\varepsilon) = (1+\varepsilon)\log(1+\varepsilon) - \varepsilon$ ,  $P(b) = (b-1)q^2x^{b-2}$  and  $e = \lfloor (k-1)/2 \rfloor$  i.i.d. even widths  $b_1, \dots, b_e \sim P$ ,  $I = \{\iota : b_{\iota} \geq \beta\} = \{\iota_1 < \dots < \iota_r\}$ ,

$$\begin{aligned} \text{pat}_k(\sigma) &\leq x^{-(n+1)} \left( \frac{x}{1-x} \right)^{k+1} \left( \mathbb{E} \min \left( 1, \prod_{j=1}^r \frac{\min((1+\varepsilon)m(b_{\iota_j}) + j + 2, k)}{b_{\iota_j} - 1} \right) \right. \\ &\quad \left. + e n q \sum_{b \geq \beta} P(b) e^{-m(b)h(\varepsilon)} \right). \end{aligned}$$

The full proof (with the  $+j+2$  bookkeeping and the Bernoulli representation) is in `experiments/w7-slots/proof.md`. Passing to the limit as in Theorem A, with  $b = (k/\theta)B$ ,  $B \sim \Gamma(2, 1)$ ,  $\beta = \bar{\beta}k$ , the exponential rate is

$$\tau - 1 - \log \tau + \log \lambda + \max(R_{\text{good}}, R_{\text{bad}}), \quad R_{\text{bad}} = -\bar{\beta}(1 - e^{-\tau/\bar{\beta}})h(\varepsilon),$$

$$R_{\text{good}} = \max_{s \geq 0} \left[ -\frac{\bar{c}s^2}{2} + \frac{1}{2} \log \left( \Pr(B < \theta\bar{\beta}) + e^s(1+\varepsilon) \mathbb{E}[(1 - e^{-\tau\theta/B}); B \geq \theta\bar{\beta}] \right) \right], \quad \bar{c} = (1+\varepsilon)\bar{\beta}(1 - e^{-\tau/\bar{\beta}}),$$

where the  $s$ -maximization accounts for the  $+j$  terms. Optimizing ( $\tau = 0.977$ ,  $\bar{\beta} = 0.594$  — i.e. one drops even indices whose window exceeds  $0.59k$ , far more aggressive than Theorem A's  $b > k + 1$  — and  $\varepsilon = 0.149$ ) the root is

$$\lambda_C = 1.00483,$$

computed independently twice (the second implementation brackets the root in  $(1.0048, 1.005)$ ).

**Corollary.**  $\text{sp}(k) \geq (1.00483 - o(1))k^2/e^2$ .

This is fifteen times the excess of Theorem A over  $1/e^2$ , and 63 times that of [2]. The finite- $k$  inequality of Theorem C, evaluated exactly (dynamic programme over  $r$ , all truncation tails added), gives  $\frac{1}{k} \log(\text{RHS}/k!) = -4.0 \times 10^{-4}$  at  $k = 30001$  and  $-6.8 \times 10^{-4}$  at  $k = 100001$  for  $\lambda = 1.004$  (it is positive at  $k = 10^4$  because of the polynomial union-bound factor). The numerics are floating point, not interval arithmetic. Why the naive route fails and this one does not: the slot counts of different windows are increasing functions of a common random set, so FKG gives

$\mathbb{E} \prod N_i \geq \prod \mathbb{E} N_i$  — the wrong direction — and negative association fails for nested value windows; but one never needs the product to factor, only a simultaneous *typical* upper bound (single-window Chernoff plus a union bound), and the price is the slack  $\varepsilon$  (the  $\varepsilon = 0$  heuristic ceiling of the same computation is 1.0117).

## 2.5 Further refinements

*Both parities.* With the stable rule  $I_{\text{even}} = \{i \text{ even} : b_i - 1 > k\}$ ,  $I_{\text{odd}} = \{i \text{ odd} : b_i - 1 > k, i \pm 1 \notin I_{\text{even}}\}$ , Lemma 1 still applies; the odd factors are 2-dependent, and a Hölder/Cauchy–Schwarz split (exponent  $p = 1.6$ ) gives  $\lambda_B = 1.000384$ , rigorous except for the routine geometric-to-exponential limit which we have not written out in full. The numerically exact rate of this two-parity encoding (transfer operator, confirmed by Monte Carlo to  $5 \times 10^{-7}$ ) is  $\lambda = 1.000437$ ; certifying it would require a verified top eigenvalue of a two-dimensional positive transfer operator. We regard  $\approx 1.0005$  as the ceiling of the “one value instead of one index” mechanism.

*What did not help.* Allowing runs of adjacent dropped indices never pays (the second index would need a window of more than  $2k$  points); optimizing a threshold is pointless since the pointwise-optimal rule is exactly “include  $i$  iff  $b_i - 1 > k$ ”.

## 3 Short superpatterns for $k = 7$ and $k = 8$

**Theorem 2.**  $\text{sp}(7) \leq 23$  and  $\text{sp}(8) \leq 30$ . Witnesses:  $\sigma_7 = 7\ 20\ 13\ 10\ 2\ 18\ 23\ 4\ 12\ 16\ 8\ 5\ 19\ 15\ 1\ 9\ 22\ 14\ 6\ 17\ 11\ 3\ 21$ ,

$\sigma_8 = 13\ 4\ 25\ 18\ 8\ 30\ 12\ 22\ 1\ 28\ 19\ 10\ 6\ 14\ 24\ 27\ 3\ 16\ 21\ 11\ 5\ 20\ 29\ 15\ 7\ 23\ 2\ 17\ 26\ 9$ .

Two further 7-superpatterns of length 23:  $7\ 18\ 12\ 2\ 15\ 23\ 6\ 19\ 13\ 4\ 8\ 11\ 22\ 17\ 3\ 20\ 10\ 5\ 14\ 1\ 21\ 9\ 16$  and  $10\ 16\ 2\ 22\ 6\ 11\ 19\ 15\ 5\ 21\ 12\ 1\ 9\ 18\ 4\ 13\ 23\ 7\ 17\ 3\ 14\ 20\ 8$ .

*How found.* Simulated annealing over  $S_n$  with moves = adjacent-position and adjacent-value swaps, energy = #missing patterns +  $0.4 \cdot \# \text{patterns realized exactly once}$ , with incremental re-evaluation (only subsets containing a swapped position change). Length 24 for  $k = 7$  was found in under a second by essentially every chain; length 23 within a minute in 3 of 4 chains; length 22 was not found (12 chains, best 9 missing patterns after  $2 \times 10^5$  moves each). For  $k = 8$ , length 32 was found after 675 moves length 31 after 3087 moves (about a minute), and length 30 with a second search variant (dynamic weights on persistently missing patterns plus occasional insertion moves); length 29 was not found (best 12 missing). An 8-superpattern of length 31 is  $25\ 21\ 11\ 4\ 27\ 8\ 17\ 19\ 1\ 13\ 31\ 5\ 23\ 16\ 26\ 10\ 6\ 29\ 15\ 20\ 28\ 3\ 9\ 14\ 24\ 2\ 12\ 30\ 18\ 7\ 22$ , and one of length 32 is  $13\ 26\ 3\ 31\ 11\ 16\ 23\ 8\ 20\ 12\ 30\ 6\ 18\ 27\ 7\ 1\ 21\ 29\ 15\ 4\ 9\ 22\ 32\ 19\ 2\ 14\ 24\ 10\ 28\ 5\ 17\ 25$ .

*Verification.* (i) Independent brute force in Python (enumerate all  $\binom{23}{7} = 245,157$  resp.  $\binom{30}{8} = 5,852,925$  subsets, collect patterns, compare count to  $k!$ ): both witnesses contain all 5040 resp. 40320 patterns. (ii) Lean 4: see §6.

*Structure.* Every solution found has LIS = LDS =  $k$  exactly (the forced minimum), short irregular runs (lengths 1–4), and no visible zigzag or grid structure; the solutions are not “word plus tie-breaking” objects. Indeed  $\zeta_k$  is *irredundant*: for  $k = 5, 6, 7$  deleting any single entry of the Engen–Vatter permutation destroys universality, so short superpatterns are not trimmed zigzags. The patterns that resist longest at  $n = 22$  ( $k = 7$ ) are near-monotone or layered (1234567, 2134567, 2176543, 7123456, ...), consistent with the analysis in [3] where layered permutations are the hard case.

*Circular superpatterns.* Manjunath and D’Souza [9] define circular  $k$ -superpatterns (patterns up to rotation) and prove  $L_{\text{circ}}(k) \leq \text{sp}(k-1) + 1$  together with a zigzag construction of length  $\lceil((k-1)^2 + 1)/2\rceil + 1$  for odd  $k$ . Theorem 2 therefore gives  $L_{\text{circ}}(8) \leq 24$  and  $L_{\text{circ}}(9) \leq 31$ , improving their 25 and 34. Their closed form for Miller’s score function is equivalent to the run-change identity used in §4 to show that zigzag-type constructions cannot beat  $k^2/2$ .

*Context.* We now have  $\text{sp}(6) = 17$ ,  $19 \leq \text{sp}(7) \leq 23$  and  $20 \leq \text{sp}(8) \leq 30$  (the lower bounds from  $\text{sp}(k+1) \geq \text{sp}(k) + 1$ , obtained by deleting the maximum entry of a  $(k+1)$ -superpattern and using the pattern  $\pi \oplus 1$ ), versus  $\lceil(k^2 + 1)/2\rceil = 19, 25, 33$ ; note  $30 < 32 = 8^2/2$ . Together with  $\text{sp}(6) = 17 < 18 = 6^2/2$ , this is mild evidence against the conjecture  $\text{sp}(k) \sim k^2/2$  of [10] (Hunter [5] announced, without proof,  $\text{sp}(k) \leq \frac{15}{32}k^2 + O(k)$ ).

## 4 Structured constructions and what cannot beat $k^2/2$

Engen–Vatter’s construction is a word over  $[k+1]$  (the zigzag) whose ties are broken so that equal letters form a decreasing subsequence. Their  $\zeta_k$  is *irredundant*: for  $k = 5, 6, 7$  no single entry can be deleted without losing a pattern. Nevertheless the tie-breaking idea is much more general than the zigzag:

**Observation.** Arnarson’s record 6-superpattern of length 17 is exactly the decreasing-tie-break of the word  $3\ 6\ 5\ 1\ 6\ 7 \mid 3\ 4\ 2\ 6\ 5\ 7 \mid 1\ 4\ 6\ 3\ 7$  over  $[7] = [k+1]$ : three near-permutation blocks with non-uniform letter frequencies. (Its inverse splits into 7 decreasing chains of consecutive values.)

Searching this family (“ $B$  distinct blocks over  $[k+1]$ , decreasing ties”) gives, all verified by two independent checkers:

- a 6-superpattern of length 18 of the form  $\rho\rho$  with  $\rho = 1\ 4\ 8\ 5\ 7\ 3\ 9\ 2\ 6 \in S_9$ ;
- a structured 7-superpattern of length  $24 = (k^2 - 1)/2$ : word  $3\ 7\ 1\ 5\ 2\ 8\ 4\ 6 \mid 2\ 4\ 8\ 1\ 5\ 3\ 6\ 7 \mid 2\ 4\ 7\ 5\ 1\ 8\ 3\ 6$ , permutation  $9\ 2\ 1\ 3\ 15\ 6\ 24\ 12\ 18\ 5\ 11\ 23\ 2\ 14\ 8\ 17\ 20\ 4\ 10\ 19\ 13\ 1\ 22\ 7\ 16$ . Three *identical* blocks  $\rho^3$ ,  $\rho \in S_8$ , never work (exhaustive: at most 5039/5040, always missing 7654321 — the same even-length phenomenon as in [3]).

For  $k = 8$ , three blocks over  $[10]$  (length 30) reach 40316/40320, missing only 54367812, 78123456, 78123546, 87654132; three blocks over  $[9]$  (length 27) fail badly. The unstructured witnesses  $\sigma_7, \sigma_8$  of §3 are **not** in this family: their minimal “row alphabet” (over all eight symmetries) is 9–10 for  $k = 7$  and 13 for  $k = 8$ , with mixed tie directions. So the family is a useful search space but the true optima leave it.

**Negative results (elementary; proofs in the project log).** (N1) Any “copy model”, in which each value of the pattern has a fixed candidate set of letters, collapses to a word over  $[k]$  containing all permutations literally and hence has length  $\geq k^2 - o(k^2)$  by Kleitman–Kwiatkowski. (N2) If the words  $\sigma|_Y$  ( $Y$  a  $k$ -set of values) take only polynomially many distinct values, Hunter’s bound on  $F(k, n)$  forces length  $\geq (1/2 - o(1))k^2$ ; so any construction whose freedom is a threshold split or a bounded number of shifts cannot beat  $1/2$ . (N3) Every zigzag-type word (monotone runs over residue classes, any alphabet, any run directions, mod- $q$  variants, with or without merged runs) needs  $\geq k - 1$  run changes to embed the alternating permutation  $1\ 3\ 2\ 5\ 4\ \dots$ , hence length  $\geq (1/2 - o(1))k^2$ ; merging two zigzags on low/high values reproduces the automaton analysed by Hunter [5] and gains only  $O(k^{3/2})$ . The common reason: gains must serve the alternating permutation ( $k - 2$  extrema) and the near-monotone permutations simultaneously, and realizations by shifts and monotone-row merges cannot do both. One might hope that rows (value classes) whose

internal tie-break pattern is non-monotone could help. For identical blocks this is provably not so: in the family  $\rho^B$  over an alphabet of size  $m = k + d$  with arbitrary row patterns, a merged value costs a block exactly like a descent, so the identity and the decreasing permutation cost  $k - \max_S \text{asc}(S)$  and  $k - \max_S \text{desc}(S)$  blocks, both maxima are attained at  $S = [m]$ , and  $\text{asc}(\rho) + \text{desc}(\rho) = m - 1$ ; hence  $B \geq (k - d + 1)/2$  and the length is at least  $(k - d + 1)(k + d)/2$ , i.e.  $(1/2 - o(1))k^2$  whenever  $d = o(k)$ . Consistently, the record witnesses embed every hard family (monotone, alternating, layered) using monotone fibres only; their few non-monotone rows are single-element displacements that mop up irregular patterns. Whether *non-identical* blocks, or alphabets  $(1 + \delta)k$  with  $\delta = \Theta(1)$ , can beat  $1/2$  remains open.

## 5 Exact values for words over $[k + 1]$

Let  $f(k; r)$  be the least length of a word over  $[r]$  containing every  $\pi \in S_k$  as an order-isomorphic subsequence;  $f(k; r) \geq \text{sp}(k)$ , and  $f(k; k + 1) \leq (k^2 + k)/2$  [4].

**Theorem 3.**  $f(2; 3) = 3$ ,  $f(3; 4) = 6$ ,  $f(4; 5) = 10$ ,  $f(5; 6) = 15$ ; i.e.  $f(k; k + 1) = (k^2 + k)/2$  for  $2 \leq k \leq 5$ . Moreover  $f(3; 5) = f(3; 6) = 5 = \text{sp}(3)$ ,  $f(4; 6) = f(4; 7) = 9 = \text{sp}(4)$ , and  $f(5; 7) \geq 13 = \text{sp}(5)$ .

Witnesses: 232; 241323; 2435142314; 135642135642135 (Miller’s word); 25314; 314625314; 371426315.

*Method.* SAT encoding (CaDiCaL via PySAT): one-hot letter variables  $x_{i,\ell}$ ; for every target word  $t = s \circ \pi$  ( $\pi \in S_k$ ,  $s : [k] \rightarrow [r]$  increasing) reachability variables  $R_{i,j}$  (“ $t_{1..j}$  embeds in  $w_{1..i}$ ”) with clauses  $R_{i,j} \rightarrow R_{i-1,j} \vee (R_{i-1,j-1} \wedge x_{i,t_j})$ , and one clause per  $\pi$  requiring  $\bigvee_s R_{n,k}$ . Symmetry breaking by complementation ( $w_1 \leq \lceil r/2 \rceil$ ). Lower bounds are UNSAT results ( $n = 13$ : 19 s,  $n = 14$ : 166 s for  $k = 5, r = 6$ ); every SAT witness was re-verified by an independent greedy embedding checker; the encoding reproduces Newey’s  $f(3; 3) = 7$ ,  $f(4; 4) = 12$ .

*Remarks.* This answers Hunter’s Problem 3 affirmatively for  $k \leq 5$ . Note  $f(k; k + 1) > \text{sp}(k)$  strictly for  $k = 3, 4, 5$ , while one more letter already recovers  $\text{sp}(k)$  for  $k = 3, 4$ . The counting bound of [3] is slack at length  $(k^2 + k)/2 - 1$ , so the UNSAT proofs are genuinely structural.  $f(6; 7) \in [17, 21]$  remains open (the  $n = 17$  instance did not finish).

## 6 Gupta’s rosaries

A *rosary* for  $n$  is a cyclic word over  $[n]$  such that every  $\pi \in S_n$  occurs as a subsequence of some rotation read once around;  $r(n)$  is the least length. Gupta conjectured  $r(n) \leq n^2/2$  [11]; Lecouturier–Zmiaikou proved it for even  $n$  and showed  $r(n) < n^2/2 + n/4 - 1$  for odd  $n$  [6].

**Theorem 4.**  $r(2) = 2$ ,  $r(3) = 4$ ,  $r(4) = 8$ ,  $r(5) = 12$  (witnesses 12, 1232, 12341232, 123432152435);  $r(6) \in \{16, 17\}$ ; and for odd  $n \in \{7, 9, 11\}$  the word

$$R_n = (1\,2\,\dots\,n)^{\lfloor m/2 \rfloor} (1\,2\,\dots\,m+1) (n\,\dots\,2\,1)^{\lceil m/2 \rceil - 1} (n\,\dots\,2), \quad m = \frac{n-1}{2},$$

of length exactly  $\lfloor n^2/2 \rfloor$ , is a rosary. Consequently Gupta’s conjecture holds for all  $n \leq 11$ .

*Method.* Exact values by SAT on the cyclic word (start position carries  $\pi(1)$ ; symmetry breaking by relabeling and rotation);  $r(6) \geq 16$  from UNSAT at 14, 15.  $R_n$  is the word of [6, Thm 2] with its partial block shortened; verified by an independent C checker (DFS over permutation prefixes carrying the live set of start positions;  $n = 11$ : 7 s).

*Dead end, recorded deliberately.*  $R_n$  is **not** a proof route: [6] state that this word fails at  $n = 21$  and exhibit a permutation for  $n = 33$ ; we re-verified that their permutation is not contained in  $R_{33}$ . The hard permutations are near-alternating with all decreasing-block minima above  $(n + 1)/2$ . A proof for odd  $n$  needs a different construction.

## 7 Random permutations (data)

Let  $t(n)$  be the least  $m$  such that at least half of  $S_m$  are  $n$ -superpatterns [1]; [3] report  $t(n) = 1, 3, 7, 13, 20, 28, 36, 48$  for  $n \leq 8$ . Our sampling (300 permutations per length) gives, for  $n = 8$ : fraction of superpatterns 0.14 at  $m = 44$ , 0.59 at 48, 0.91 at 53, consistent with  $t(8) = 48$ . For  $n = 9$  (60 samples per length, fixed seed) the fractions are 0.43, 0.53, 0.75, 0.80 at  $m = 58, 60, 62, 64$ , giving  $t(9) \approx 60$ , and 20 samples per length for  $n = 10$  give  $t(10) \approx 70$ –72 (new values; we also find  $t(7) = 37$  rather than the sampled 36 of [3]:  $0.41 \pm 0.02$  at  $m = 36$  versus 0.55 at  $m = 37$  with 1000 samples each;  $t(9)/81 \approx 0.74$ , in line with  $t(n) \approx 0.75n^2$  over this range and far above  $n^2/4$ ). The patterns that appear *last* near threshold are not monotone: for  $n = 8$ ,  $m = 44$  the most frequently missing patterns are 1 3 2 6 4 8 7 5, 1 3 2 4 8 5 6 7, 2 1 4 3 5 6 7 8, 1 3 2 5 4 7 6 8 — identities perturbed by a few adjacent transpositions. This is consistent with the analysis of He–Kwan’s proof [7], where the  $\log \log k$  loss comes entirely from the union bound over “structured maps” (unions of few increasing runs), and suggests these hybrid patterns are the real obstacle to Alon’s conjecture  $t(n) \sim n^2/4$ . The same ordering of difficulty (indecomposable < monotone < extremal) appears, in a different model, in the online-embedding results of Altschuler, Dubroff and Tikhomirov [12].

*A falsifiable test of Alon’s conjecture.* For  $\tau \in S_j$  let  $L_\tau(N)$  be the largest  $L$  such that  $\tau^{\oplus L}$  (the direct sum of  $L$  copies) is contained in  $N$  uniform random points, and  $c_\tau = \lim L_\tau(N)/\sqrt{N}$  (which exists by superadditivity). Alon’s conjecture implies  $c_\tau \geq 2/j$  for every  $\tau$ : otherwise the pattern  $\tau^{\oplus L}$  of length  $k = jL$  would need more than  $(1/4 + \varepsilon)k^2$  points. Monotone  $\tau$  have  $c_\tau = 2/j$  exactly (from LIS  $\sim 2\sqrt{N}$ ), and in general  $c_\tau \leq \min(2/\text{lis}(\tau), 2\text{lds}(\tau)/j, e/j)$  (Greene’s theorem with Baik–Deift–Johansson for the first two bounds, a first-moment count for the third). Using an exact dynamic programme for  $L_\tau(N)$  (validated against brute force on 1824 point sets  $\times$  152 patterns with no mismatch), we estimated  $c_\tau$  for all 64 symmetry classes of  $\tau$  of length 2 to 5 at  $N$  up to  $10^5$ . The paired statistic  $jL_\tau - \text{LIS}$  on the same point set has mean in  $[-5.3, 2.3]$  at  $N = 25000$  (LIS  $\approx 300$ ) for every  $\tau$ , i.e.  $c_\tau = 2/j$  to within about 0.01, with the  $z$ -scores of the 58 best-sampled classes spread exactly as noise. **There is no counterexample candidate**, and the data support the sharper statement  $c_\tau = 2/j$  for all  $\tau$  — equivalently, that a “ $\tau$ -Hammersley” growth process has the flux relation of the Hammersley process at intensity  $N/j^2$ . We could not prove this: the known proofs of the LIS constant rely on exact solvability. Finally, for the hybrid patterns of length  $k = 8, \dots, 32$  (identity with spaced adjacent transpositions, and  $(21)^{\oplus k/2}$ ) the half-probability containment threshold equals that of the identity within  $\pm 3\%$ , and  $N_{\text{id}}/(k^2/4) = 1.57, 1.43, 1.38, 1.34, 1.30, 1.23$  for  $k = 8, 12, 16, 20, 24, 32$ , decreasing like the Tracy–Widom  $N^{1/6}$  correction — consistent with the ratio tending to 1.

### 7.1 Rigorous bounds on $c_{21}$ , and a universal absence threshold

The first open case of the conjecture  $c_\tau = 2/|\tau|$  is  $\tau = 21$ : is the layered pattern  $2\ 1\ 4\ 3 \dots (2L)(2L-1)$  contained in  $(1 + \varepsilon)L^2$  random points w.h.p.? Previously known:  $0.40 \leq c_{21} \leq e/2$  (a diagonal greedy and the first moment). We prove  $0.787 \leq c_{21} \leq 1.140$ .

*Lower bound.* Tile the square with cells of side  $\delta/\sqrt{N}$ ; cells containing a copy of  $\tau^{\oplus t}$  are i.i.d. Bernoulli( $p_{\tau,t}(\delta^2)$ ), and a strictly increasing chain of such cells is a  $\tau$ -chain. The strictly increasing

chain in i.i.d. Bernoulli( $p$ ) sites on an  $n \times n$  grid has length  $\sim n \cdot 2\sqrt{p}/(1 + \sqrt{p})$  (Seppäläinen 1997), so  $c_\tau \geq \sup_{\delta, t} t g(p_{\tau, t}(\delta^2))/\delta$ . For  $\tau = 21$ ,  $t = 1$  ( $p = 1 - e^{-\lambda} I_0(2\sqrt{\lambda})$ ,  $\lambda = \delta^2$ ) gives 0.577 at  $\delta = 1.11$ , and  $t = 2$  (cells containing 2143, exact counts) gives 0.598. The same construction gives  $c_\tau \geq 0.55\text{--}0.60 \cdot (2/|\tau|)$  for all  $\tau$  of length  $\leq 5$ , and  $c_\tau \geq (0.7 - o(1))/(|\tau|\sqrt{\ln|\tau|})$  in general; it cannot exceed  $0.6 \cdot 2/|\tau|$  by construction (one block per cell).

*Better lower bound: a renewal sweep.* Work with a unit-intensity Poisson process on the quadrant. From a corner  $c$  sweep the squares  $c + (0, s)^2$  with  $s$  increasing and stop at the first arrival that completes a copy of 21 inside the swept square; the earlier points then form an increasing chain, the new point  $b$  lies on the top or right edge, and we take as partner the chain point  $a$  with the least coordinate that forms a descent with  $b$ . The corner of the copy,  $c' = \text{coordinatewise max of } \{a, b\}$ , dominates  $b$ , and since  $b$  is on the boundary of the open swept square, the quadrant above-right of  $c'$  is disjoint from the swept region. Hence the increments  $(dx, dy) = c' - c$  of successive steps are i.i.d., and in  $N$  points of the unit square the chain of copies has length  $(2 + o(1))\sqrt{N}/\mathbb{E}[dx + dy]$ , i.e.  $c_{21} \geq 2/\mathbb{E}[dx + dy]$  (this is Hammersley's 1972 argument for the LIS, applied to descent pairs). Conditioned on the stopping radius  $r$  and on  $n$  earlier points, those points are i.i.d. uniform in  $(0, r)^2$  conditioned to be a chain, so their sorted  $x$ - and  $y$ -coordinates are independent uniform order statistics,  $P(n) = n/(n+1)!$ ,  $\mathbb{E}r = \Gamma(n + \frac{3}{2})/n!$ , and the cost  $dx + dy = r + x_a$  has the exact series

$$\mathbb{E}[dx + dy] = \sum_{n \geq 1} \frac{\Gamma(n + \frac{3}{2})}{n! (n+1)!} \sum_{k=0}^{n-1} \left(1 + \frac{k+2}{n+2}\right) = 2.5425687 \dots,$$

giving  $c_{21} \geq 0.7866$ . (Series recomputed and the rule simulated independently:  $2.5434 \pm 0.002$  from  $4 \cdot 10^5$  steps. A strip-shaped sweep gives the closed form  $\sqrt{2e(4-e)} = 2.6397$ , i.e. 0.7577; sweeping  $\ell_{1.5}$ -balls gives 0.803 in simulation but no closed form.) The same sweep works for any  $\tau \in S_j$ : conditioned on the stopping radius the earlier points are a uniform  $\tau$ -avoiding permutation, so  $\mathbb{E}[dx + dy]$  is an exact series in  $|\text{Av}_m(\tau)|$  and first-occurrence counts (enumerated for  $m \leq 11$ , tails bounded via the Catalan, Gessel and Bóna formulas). This gives  $c_\tau \geq 0.535\text{--}0.552$  for all  $\tau \in S_3$  (cell construction: 0.389; conjecture  $2/3$ ) and  $c_\tau \geq 0.385\text{--}0.413$  for all  $\tau \in S_4$  (cell construction: 0.30; conjecture  $1/2$ ); the square-rule cost was re-simulated independently for 123, 231 and 21 and agrees with the series to 0.3%. The method is intrinsically capped: any renewal rule pays at least the cheapest descent pair in the quadrant, whose mean cost is 2.386 (simulation), so it cannot exceed  $c_{21} \approx 0.84$ ; the remaining gap to the conjectured 1 is the same 0.80 ratio by which Hammersley's greedy ( $\sqrt{8/\pi}$ ) misses the LIS constant 2, and closing it needs correlation between consecutive blocks rather than a better local rule.

*Upper bound: canonical copies.* Among all copies of  $\pi \in S_k$  in  $\Pi_N$  take one minimizing  $\sum_r x_r$ ; then for every  $r$  the strip  $S_r = (x_{r-1}, x_r) \times (y_{(\pi(r)-1)}, y_{(\pi(r)+1)})$  contains no point (else moving  $p_r$  there gives a copy with smaller  $\sum x$ ), and the strips are pairwise disjoint. By the Mecke formula,

$$\mathbb{E}\#\{\text{canonical copies}\} = N^k \int_{\Delta_x \times \Delta_y} \exp\left(-N \sum_r g_r(h_{\pi(r)-1} + h_{\pi(r)})\right) dg dh,$$

in  $x$ -gap and  $y$ -gap coordinates, and permuting the  $x$ -gaps shows that **this quantity is the same for every**  $\pi \in S_k$  (it refines the identity  $\mathbb{E}\#\text{copies} = \binom{N}{k}/k!$ ). A Chernoff extension of both simplices to orthants, integration of the  $x$ -gaps, and the scaling  $h = u/\sqrt{N}$ ,  $s = t = \sigma\sqrt{N}$  leave a one-dimensional positive transfer operator  $(T_\sigma f)(u) = \int_0^\infty e^{-\sigma u'} f(u')/(\sigma + u + u') du'$  and  $\mathbb{E} \leq N^{-1/2} e^{2\sigma\sqrt{N}} \int e^{-\sigma u_0} (T_\sigma^k 1)(u_0) du_0$ . A Collatz–Wielandt test function ( $\sigma = 1.5$ ,  $\varphi(u) = 10^{-3} +$

$(1 + u/1.7777)^{-0.9584}$ ,  $\sup T\varphi/\varphi \leq 0.26808$ , recomputed independently) gives, for **every**  $\pi \in S_k$ ,

$$\Pr(\pi \subseteq \Pi_N) \leq 668 N^{-1/2} \exp(3\sqrt{N} - 1.3163 k).$$

Hence every pattern of length  $k > 2.279\sqrt{N}$  is absent w.h.p. (the first moment gives  $e\sqrt{N} = 2.718\sqrt{N}$ ),  $c_\tau \leq 2.279/|\tau|$  for all  $\tau$ ,  $c_{21} \leq 1.140$ , and the containment threshold of *every individual*  $\pi \in S_k$  is at least  $(0.1925 - o(1))k^2 = (1.42/e^2)k^2$ , versus  $k^2/e^2$  from counting. For the identity the same bound says  $\text{LIS} \leq 2.279\sqrt{N}$  (truth 2), so the method is about 14% off in both cases; canonical rules tailored to the 21 structure did worse than the universal one. (This does not give a deterministic superpattern bound: the number of canonical  $k$ -subsets of a fixed permutation is not controlled.) The spectral-radius certificate is floating-point quadrature, not interval arithmetic; the rest is exact.

*Is the identity the easiest pattern?* Numerically it is the *hardest* at finite  $k$ : the half-probability containment threshold  $n_{1/2}(\pi)$  of a random  $\pi \in S_k$  is 3–6% below that of the identity for every  $k \leq 16$  tested (e.g.  $k = 8$ ,  $n = 26$ : the identity is contained with probability 0.55, random patterns with 0.60–0.62; recomputed independently). In units of  $k^2$  the thresholds decrease with  $k$  — identity 0.405, 0.373, 0.360, 0.345, 0.341, 0.339 and random 0.384, 0.362, 0.348, 0.334, 0.321, 0.304 for  $k = 8, \dots, 20$  (the  $k = 20$  values rest on 120 samples,  $\pm 3\%$ ) — and a fit  $n_{1/2} = (k/2 + ak^{1/3})^2$  is consistent with both tending to  $k^2/4$  (i.e.  $\kappa_{\text{univ}} = 2$ ) with a smaller finite-size correction for random  $\pi$ , but a random-pattern limit of  $0.23k^2$  fits equally well, and the ratio random/identity (0.95, 0.97, 0.97, 0.97, 0.94, 0.90) is if anything widening. If the random limit is indeed  $\approx 0.23k^2$  then  $\kappa_{\text{univ}} \approx 2.09 > 2$ , the identity is *not* asymptotically the easiest pattern, and  $c_\tau \leq 2/|\tau|$  cannot come from any  $\pi$ -uniform bound. Runs at  $k = 24$ – $32$  with  $\geq 500$  samples would decide; the question is open. On the theory side, a second level of the lex-min hierarchy (two-point moves, which add constraints only for rank-adjacent pairs) lowers the identity’s constant to  $\approx 2.24$  (transfer computation, uncertified) and mixed leftmost/lowest rules are strictly worse (2.35), so the universal constant 2.279 stands.

## 7.2 Towards Alon’s conjecture: a larger class at $n = O(k^2)$

He and Kwan [7] prove that a random permutation of length  $2000k^2 \log \log k$  is a  $k$ -superpattern, and that length  $20k^2$  suffices for the *quasirandom* patterns  $Q_k = \{\pi : L_\Delta(\pi) \leq 3\sqrt{k} \text{ for all } \Delta\}$ , where  $L_\Delta(\pi)$  is the length of the longest  $\Delta$ -shift chain (the LIS of the partial map  $i \mapsto \pi^{-1}(\pi(i) + \Delta)$ ). Their argument reduces to a random  $2k \times m$  Bernoulli( $\frac{1}{2}$ ) matrix  $M$  ( $m = n/4k$ ) and runs  $\ell$  greedy “threads” at row shifts  $t_1 < \dots < t_\ell$ ; two threads share exposed cells only along  $\Delta$ -shift chains, and they bound the sharing by  $L_\Delta \cdot (\text{maximal zero-run in } M) \leq L_\Delta \log^2 k$ . We sharpen both halves of that bound.

**Lemma 3.** For shifts  $t < t + \Delta$ , with  $(z_a)$  the zero-run profile of the leader thread, the number of shared cells is at most  $W_\Delta(z) = \max_A \sum_{a \in A} (z_a + 1)$  over  $\Delta$ -shift chains  $A$  of elements reached by the leader.

**Theorem 7.** Let  $\mathcal{A}_{L,s}$  be the event that every set of at most  $L$  cells in distinct rows has total zero-run less than  $s$  (so  $\Pr(\neg \mathcal{A}_{L,s}) \leq 2(2km)^L e^{-s/8}$  for  $s \geq 4L$ ). If the pairwise shifts satisfy  $L_{t_j - t_i}(\pi) \leq L$ ,  $s \geq 4L$  and  $(\ell - 1)(s + L) \leq m/2$ , then

$$\Pr(M \not\supseteq \pi \text{ and } \mathcal{A}_{L,s}) \leq \Pr(\text{Bin}(\ell(m/2 - k), \tfrac{1}{2}) \leq \ell(k - 1)).$$

The proof makes He–Kwan’s device of “pretending that  $\mathcal{A}$  holds” precise: a cell is pretended to be a 1 (without exposure) exactly when exposing it as 0 would violate the run-sum condition; on  $\mathcal{A}$

every pretended cell is a genuine 1, the modified threads coincide with the real ones, and Lemma 3 (which uses only the interval structure of the threads) bounds each pairwise overlap by  $s + L$ ; the really exposed cells then number at least  $\ell(m/2 - k)$  with at most  $\ell(k - 1)$  ones, and the exposure principle applies.

**Theorem 8.** For large  $k$  a uniformly random permutation of length  $72k^2$  contains w.h.p. every  $\pi \in S_k$  for which fewer than  $k/(3 \ln k)$  shifts  $\Delta$  have  $L_\Delta(\pi) > k/(3 \ln^2 k)$ . The exceptional set has size at most  $k! e^{-(1/3 - o(1))k/\ln k}$ .

*Proof.* Take  $m = 18k$ ,  $\ell = \lceil \ln k \rceil$ ,  $L = \lfloor k/(3 \ln^2 k) \rfloor$ ,  $s = \lceil 20L \ln k \rceil$ ; then  $\Pr(\neg \mathcal{A}_{L,s}) \leq 2 \exp(L(2 \ln k + 3.6) - 2.5L \ln k) \rightarrow 0$ . For such  $\pi$  choose  $\ell$  shifts in  $\{0, \dots, k\}$  greedily so that no pairwise difference lies in the set  $F$  of bad shifts (each choice forbids  $2|F| + 1 < 2k/(3 \ln k) + 1$  values, so  $\ell$  choices fit). Theorem 7 applies since  $(\ell - 1)(s + L) \leq 7k + \ln k \leq m/2$ , and gives  $\Pr \leq \Pr(\text{Bin}(8\ell k, \frac{1}{2}) \leq \ell k) \leq e^{-8\ell k D(1/8 \| 1/2)} \leq k^{-2.5k}$ ; a union bound over  $k^\ell$  patterns finishes, and He-Kwan's coupling (their Lemma 2.2) transfers the statement from  $M$  to permutations of length  $2 \cdot 2k \cdot m = 72k^2$ . The size of the exceptional set follows from  $\Pr(\max_\Delta L_\Delta(\pi) \geq L') \leq k(e^2 k/L'^2)^{L'}$  for uniform  $\pi$  [7, Lemma 3.1]. ■

Since  $k/(3 \ln^2 k) \gg 3\sqrt{k}$ , this strictly enlarges the class handled at  $n = O(k^2)$ : the patterns *not* covered number  $k!e^{-\Theta(k/\ln k)}$  instead of  $k!e^{-\Theta(\sqrt{k})}$ , and He-Kwan's own tightness family (unions of  $\log^{10} k$  increasing runs on value intervals) now lies inside the covered class. The full proofs are in `experiments/w9-alon-threads/proof.md`; Lemma 3 was also checked by simulation (no violations in 400 instances; numerically  $W_\Delta$  is 6–8 times below He-Kwan's bound).

**Removing the  $\log k$  from the global event.** The cells that matter in Lemma 3 are the leader's *start cells* along a shift chain; their columns increase with the element, so listed by column their row sequence is  $\pi|_A$ , and if  $\text{LDS}(\pi|_A) \leq \rho$  they form a union of at most  $\rho$  monotone (row- and column-increasing) chains — a  $\rho$ -staircase set. The number of  $\rho$ -staircase sets of size  $j$  in  $[2k] \times [m]$  is at most  $\binom{2k}{j} \binom{m}{j} \rho^{2j}$  (choose rows and columns; a permutation with  $\text{LDS} \leq \rho$  is determined by labelling positions and values with their increasing-subsequence class), i.e.  $e^{O(j \log(\rho k/j))}$  rather than  $(2km)^j$ . Replacing  $\mathcal{A}_{L,s}$  in Theorem 7 by the event “every  $\rho$ -staircase set of  $\leq L$  cells has run-sum  $< s$ ” (the proof goes through verbatim) lets one take  $s = O(L \log(\rho k/L))$ .

**Theorem 5.** Let  $\mathcal{L}_r$  be the unions of at most  $r$  increasing runs on value intervals and  $\mathcal{D}_r = \{\pi : \text{LDS}(\pi) \leq r\}$ . For large  $k$ , a uniformly random permutation of length  $800k^2$  contains w.h.p. every  $\pi \in \mathcal{L}_r$  with  $r \leq k/(2 \ln(k + 1))$ , and every  $\pi \in \mathcal{D}_r$  with  $r \leq k$ , provided that fewer than  $k/(2\ell_r) - \frac{1}{2}$  shifts  $\Delta$  have  $L_\Delta(\pi) > k/\lambda_r$ , where  $\ell_r = \lceil (\ln r + 1)/29 \rceil$  (resp.  $\lceil (2 \ln r + 1)/29 \rceil$ ) and  $\lambda_r = (\ell_r - 1)(\ln r + 10) = O(\ln^2 r)$ . In particular there is **no condition at all** for  $r \leq e^{28}$ ; using the two row-disjoint shifts  $t = 0$  and  $t = k$  (a  $k$ -shift chain is empty) this extends to  $r \leq e^{57}$  for  $\mathcal{L}_r$  and  $r \leq e^{28}$  for  $\mathcal{D}_r$  with no condition.

The proof (in `experiments/w13-global-event/proof.md`) uses  $\ell_r$  threads and the staircase event with  $\rho = r$ ; the union bound over the  $\leq (k+1)^r r^k$  patterns of  $\mathcal{L}_r$  (resp.  $r^{2k}$  of  $\mathcal{D}_r$ ) is paid by  $e^{-29\ell_r k}$  per pattern. For random interleaving words the hypothesis holds with room ( $\max_\Delta L_\Delta \leq 1.5k/\sqrt{r}$  empirically), so this covers the “run-quasirandom” part of the residual class of Theorem 8, and at  $r = k$  it reproduces Theorem 8 with threshold  $14.5k/\ln^2 k$  in place of  $k/(3 \ln^2 k)$ . Numerically, riding between threads for run-unions is concentrated on shifts aligned with the run height (joint failure ratio 2.4 at  $\Delta = \text{height}$ ,  $\approx 1.0$ – $1.2$  elsewhere).

**Speed- $N$  bounds for bounded numbers of runs.** For the union-of-runs class one can bypass threads entirely. Write such a pattern as  $\pi_w$  for an interleaving word  $w \in [r]^k$  (letter  $j$  marks the

positions of run  $j$ ). Two elementary facts:  $\pi_w \subseteq P$  iff  $P$  contains  $r$  vertically *separated* chains whose merge word is  $w$ ; and every  $w$  is a subsequence of the periodic word  $(12\cdots r)^{\rho(w)}$  with  $\rho(w) = 1 + \#\{i : w(i+1) \leq w(i)\} \leq k$ , so every union of  $r$  runs is a pattern of the periodic shuffle. For the periodic shuffle itself, tile the square into  $4r^2(r+1)m^2$  boxes ( $k = rm$ ) and call a box good if it is nonempty; a “molecule” is a choice of one good box in each of  $r$  consecutive box-columns, one per strip, and a chain of  $m$  molecules in the product order yields a copy. A Mirsky-type argument (the molecule poset has at least  $n_0 n_1^r - n_1^{r-1} B$  elements and width at most  $n_1^r + r n_0 n_1^{r-1}$ ,  $B$  the number of bad boxes) shows a chain of length  $m$  exists whenever  $B \leq 2m^2$ , and a Chernoff bound gives:

**Theorem 9.** For  $r \geq 2$  and  $N \geq 8(r+1) \ln(2er^2(r+1)) k^2$ ,  $\Pr((12\cdots r)^{k/r} \not\subseteq \Pi_N) \leq e^{-N/(4r^2(r+1))}$ ; consequently, for every fixed  $r$ , a uniformly random permutation of length  $n \geq 16r^2(r+1) \ln(2er^2(r+1)) k^2$  contains every union of at most  $r$  increasing runs on value intervals of length  $k$  with probability  $1 - e^{-\Omega(n/r^3)}$ .

The failure probability has speed  $N$  (like the LIS lower tail), which is what a union bound over the  $e^{\Theta(k \log r)}$  residual patterns needs; the limitation is the threshold, linear in  $r \log r$ , so on the residual class  $r \leq \ln^4 k$  this gives only  $n = k^2 \text{polylog } k$ . A matching lower bound in the fixed-strip model ( $\Pr \geq e^{-N/r}$ ) shows that the speed is  $N/\text{poly}(r)$  genuinely. Proofs in `experiments/w11-strips/proof.md`.

**Theorem 10 (block-grid patterns: an  $r$ -independent threshold).** Let  $k = rh$  and, for  $\tau = (\tau_0, \dots, \tau_{h-1}) \in (S_r)^h$ , let  $\pi_\tau(ir + s) = \tau_i(s)h + i$  ( $0 \leq i < h$ ,  $0 \leq s < r$ ); the class  $\mathcal{G}(r, h)$  of all such  $\pi_\tau$  contains the tilted grid ( $\tau_i = \text{id}$ , i.e. the periodic word  $(12\cdots r)^h$ ) and the whole family  $\mathcal{F}(r, h, \varepsilon)$  of the previous paragraph. Let  $N = Ck^2$  and let  $c(C) = \sup_{\theta > 0} (\theta - \ln \mathbb{E} e^{\theta U})$ , where  $(U, V)$  has density  $Ce^{-C(u+v)^2/2}$  on the quadrant, so that  $\mathbb{E}U = \sqrt{\pi/(8C)}$  and  $c(C) > 0$  iff  $C > \pi/8$  ( $c(1) = 0.20$ ,  $c(2) = 0.75$ ). Then for every  $\pi \in \mathcal{G}(r, h)$

$$\Pr(\pi \not\subseteq \Pi_N) \leq e^{-c(C)k} + r e^{-c(C)h},$$

and by reflection also  $\leq e^{-c(C)k} + h e^{-c(C)r}$ . Consequently a uniformly random permutation of length  $n = (\pi/8 + \varepsilon)k^2 \approx 0.393k^2$  contains every  $\pi \in \mathcal{G}(r, h)$  with probability  $1 - o(1)$  as  $\min(r, h) \rightarrow \infty$ , uniformly in  $\pi$ ; and at  $n = k^2$  every block-grid pattern with  $r \leq k/(6 \ln k)$  is contained with probability  $\geq 1 - (r+1)k^{-1.2}$ .

*Proof sketch (full proof: `experiments/w11-strips/proof.md` §4).* Cut the values into  $r$  strips of  $h$  consecutive values and the positions into  $h$  slabs of  $r$ ;  $\pi_\tau$  puts exactly one point in each (slab, strip) cell, visiting the strips of slab  $i$  in the order  $\tau_i$  and, within a strip, ordered by slab. Complete each strip’s points by an independent Poisson process outside the strip, so the  $r$  strip-processes are independent Poisson processes on the plane. Now embed greedily in position order: keep a global  $x$ -level  $a$  and a  $y$ -level  $b_j$  per strip, and at the cell  $(i, \tau_i(s))$  take the point of the strip-process in the quadrant  $\{x > a, y > b_j\}$  minimising  $(x - a) + (y - b_j)$ ; advance both levels to it. By the Mecke formula the scaled increments  $k(x - a), k(y - b_j)$  have, conditionally on the past, the fixed law  $(U, V)$  above (the explored triangles of earlier searches in the same strip are disjoint from the current quadrant, so the process there is still Poisson); hence all  $k$  increment pairs are i.i.d. On the event that the  $k$   $x$ -increments sum to less than  $k$  and, in each strip, the  $h$   $y$ -increments sum to less than  $h$ , every chosen point lies in its own cell (so it is a genuine point of  $\Pi_N$ ) and the chosen points form a copy of  $\pi_\tau$  by construction; Chernoff bounds the complement. The rule  $(x - a) + (y - b_j)$  is optimal among corner rules minimising a 1-homogeneous function (squares need  $C > 9\pi/64$ , quarter-discs  $C > 4/\pi^2$ ). De-Poissonisation is the standard coupling of  $\Pi_{(1-\delta)n}$  inside  $n$  uniform points. ■

*Discussion.* The greedy uses nothing about  $\tau$  beyond bijectivity of each  $\tau_i$ , so it is insensitive to exactly the structure (rows visiting strips in arbitrary orders) that made  $\mathcal{F}$  defeat every chain-based hypothesis. Its threshold  $\pi k^2/8$  is  $r$ -independent and below the *rigid-slab* threshold  $k^2$  of the identity itself; the true threshold of these patterns is conjecturally  $k^2/4$ , so the loss is a factor  $\pi/2$ , the price of taking increments one cell at a time. What Theorem 10 does not give is a union bound:  $|\mathcal{F}| = e^{\Theta(k \ln k)}$  while the failure probability has speed only  $\min(r, h)$  in the rows (one bad row kills the greedy) — Theorem 4.4 of the proof file improves the row part to speed  $k$ , but the column part stays  $re^{-c(C)h}$ . So  $\mathcal{F}$  is *not* a counterexample to Alon’s conjecture, and the obstruction it poses is to the method (per-pattern bounds summed over shift-chain classes), not to the statement. We checked the theorem numerically: the greedy, run with an assertion that the chosen points form  $\pi_r$ , succeeds in 86% of trials at  $r = h = 10$ ,  $C = 1$  and never produced a non-copy. Exact threshold computations (a Pareto-front dynamic program for block-grid containment, validated against brute force on 7200 samples) show the fixed-strip threshold of the  $r \times r$  tilted grid *decreasing* with  $r$ :  $N_{1/2}/k^2 = 0.55, 0.50, 0.44, 0.41, 0.375, 0.36$  for  $k = 4, 9, \dots, 49$ , about 0.06 above the identity’s finite-size values, and in the free model every tilted grid with  $k \leq 32$  is *easier* than the identity of the same length.

**Theorem 11 (every pattern below  $k^2$ : the reserve greedy).** For every  $k$  and every  $\pi \in S_k$ , with  $N = Ck^2$ : (a) (rigid rows)  $\Pr(\pi \not\subseteq \Pi_N) \leq \Pr(\text{Poisson}(Ck) \leq k-1) \leq e^{-k(C-1-\ln C)}$ , so the universal threshold is at most  $k^2$  and a uniformly random permutation of length  $(1+o(1))k^2 \ln k$  is a  $k$ -superpattern (the union bound with an explicit constant; He–Kwan’s threads in Poisson form). (b) Let  $C_{\text{univ}}$  be the threshold of the two-point reserve chain defined below; numerically  $C_{\text{univ}} = 0.757$ . For every  $C > C_{\text{univ}}$  there is  $\eta(C) > 0$  ( $\eta(0.8) \geq 0.0014$ ,  $\eta(1) \geq 0.035$ ,  $\eta(1.2) \geq 0.096$ ) such that  $\Pr(\pi \not\subseteq \Pi_{Ck^2}) \leq e^{-\eta(C)k+O(1)}$  for **every**  $\pi \in S_k$ ; hence a random permutation of length  $(0.757 + \varepsilon)k^2$  contains any fixed  $k$ -pattern w.h.p.

*Proof sketch (full proof: [experiments/w19-general-greedy/proof.md](#) §2, §4B).* (a) Cut the values into  $k$  rows of height  $1/k$ ; embed the positions in order, each time taking the leftmost point of row  $\pi(p)$  to the right of the current  $x$ -level. Rows are disjoint, so the scaled  $x$ -increments are i.i.d.  $\text{Exp}(C)$  and the embedding fits iff  $\text{Poisson}(Ck) \geq k$ . (b) Cut the value sequence  $\pi^{-1}(1), \pi^{-1}(2), \dots$  into maximal monotone runs and split them into pieces of length 2 and 3 (any two consecutive values are monotone, so this is always possible). Each piece is a strip of height  $2/k$  or  $3/k$  on which the positions are monotone. Embed in position order; for a strip with  $m \geq 2$  points still to place, search *inside the strip* above the current level but keeping a **reserve** of  $\beta$  rows per future point, taking the point minimising  $(x-a) + \lambda(y-b)$ ; the last point of a strip is the leftmost one in what remains. The strip’s own future depends on the past only through the height left in that strip, and different strips search independent (extended) Poisson processes, so the total  $x$ -cost is a sum of independent strip costs  $T_j$  with an explicit law; a copy of  $\pi$  exists as soon as  $\sum_j T_j < k$ , and Chernoff gives  $e^{-\eta(C)k}$  whenever  $\mathbb{E}T_j < h_j$  for both strip lengths. With  $\beta = 0.4$ ,  $\lambda = 1.1$ :  $\mathbb{E}T/2 = 1.0009$  at  $C = 0.757$  and  $0.947$  at  $C = 0.8$  for  $h = 2$ ;  $\mathbb{E}T/3 = 0.89$  at  $C = 0.757$  for  $h = 3$  (our independent simulation; W19’s quadrature agrees). ■

*Discussion.* Theorem 11(b) is the first threshold below  $k^2$  valid for arbitrary patterns (He–Kwan give  $20k^2$  for quasirandom patterns and  $O(k^2 \log \log k)$  in general); the worst case is the alternating pattern  $214365\dots$ , whose value sequence consists of runs of length exactly 2. The gain over rigid rows comes from trading  $x$ -cost against the  $y$ -slack that rows waste. For structured  $\pi$  (long monotone value runs) one keeps the runs whole and the threshold decreases towards  $\pi/8$ . The constants are low-dimensional integrals, evaluated numerically and re-simulated independently; a certified evaluation is routine. **What this does not give:** the exponent is only  $\eta k$ , and W19

proves (Props. 5.2–5.3 there) that every greedy of this kind — including two-phase repairs — fails with probability  $\geq e^{-O(\min(k,h))}$ , so no such procedure can reach the  $e^{-(1+\varepsilon)k \ln k}$  per pattern that the union bound over  $S_k$  needs at  $n = O(k^2)$ ; with these tools the union bound gives exactly the trivial  $k^2 \ln k$ . A proof of  $n = O(k^2)$  must move all strips at once (a two-dimensional last-passage argument), which is where the next step lies.

**Symmetries and mixed directions.** Containment is equivariant under the dihedral group (reverse, complement, inverse) and a uniform permutation is invariant, so any class covered w.h.p. can be replaced by the union of its eight images; the shift chains of  $\pi^c, \pi^r$  are those of  $\pi$  with LDS and LIS exchanged, and those of  $\pi^{-1}$  are the “horizontal” chains of  $\pi$  (pairs at horizontal distance  $\Delta$  consistently ordered by value). The staircase count also extends to unions of  $\rho$  monotone chains of *either* direction (an extra factor  $2^{\min(\rho,j)}$ ), so Theorem 5 holds for unions of increasing *or decreasing* runs on value intervals, or on position intervals, each run’s direction chosen freely.

**Why chain-based hypotheses cannot finish the job.** Three negative results delimit the thread method as it stands. (i) The entropy statement one would need for structured interleavings is false: for two runs whose value intervals are near-translates by  $\Delta$ , the  $\Delta$ -shift chain has length  $\approx$  the run length *for every interleaving word* (the map  $v \mapsto v - \Delta$  is monotone on both runs), so  $L_\Delta$  carries no information about the word, and for  $r \geq e^{75}$  there are at least  $e^{61k}$  patterns in  $\mathcal{L}_r$  with only two mutually good shifts. (ii) More strongly, let  $k = rh$  and  $\pi_\tau(ir + s) = \tau_i(s)h + i$  for  $\tau \in (S_r)^h$  with each  $\tau_i$  having at least  $(1 - \varepsilon)r$  fixed points (the He–Kwan tilted grid with perturbed rows). For  $r, h \geq 20$ ,  $\varepsilon \leq 1/16$ , every dihedral image of every such  $\pi$  has  $L_\Delta \geq k/16$  for *every*  $\Delta \leq 0.7k$ , so no three shifts can be chosen; and there are at least  $e^{(1/32 - o(1))k \ln k}$  of them, too many for any bound of the form  $e^{-Dk}$  per pattern. Yet numerically the *actual* overlap of two threads on these patterns is  $O(1)$  cells: threads at shift  $\Delta = dh + e$  ride only when the follower’s element-lag  $er + d$  is  $\lesssim 3\sqrt{k}$ . So what fails is the bound  $|T_t \cap T_{t+\Delta}| \leq W_\Delta(z)$ , which charges every chain, not the threads. One would hope for a *lag lemma*: a follower whose elements trail the leader’s by  $\lambda$  along a chain should have to overtake by  $\approx \lambda$  columns before sharing a cell, at cost  $e^{-\Omega(\lambda^2/k)}$ , so that only  $O(\sqrt{k} \log k)$  shifts are harmful for any pattern — which would give  $n = O(k^2)$  for all of  $S_k$ . **This is false.** (iii) Let  $\pi$  be the tilted grid  $\pi(ir + s) = sh + i$  on  $k = rh$  elements,  $m = Ck$  columns, and let  $E_t$  be the failure event of the thread at shift  $t$ . For every shift  $e \leq h/4$  (element lag  $\lambda = er \in \{r, 2r, \dots, k/4\}$ , all  $\geq \sqrt{k}$  when  $r = h$ ) and  $C \geq 3$ ,  $k \geq 6$ ,

$$\Pr(E_0 \cap E_e) \geq 2(144C^2)^{-\lambda} \Pr(E_0),$$

and symmetrically for  $e \geq 3h/4$ ; hence for  $C \geq 8$  every such shift has  $\Pr(E_0 \cap E_e) \geq \Pr(E_0)^2 e^{\Omega(k)}$ . The proof prescribes  $4\lambda - 1$  cells (the leader finds its first  $2\lambda$  elements at columns  $0, \dots, 2\lambda - 1$ ; the follower spends columns  $2b, 2b + 1$  on element  $b < \lambda$ ), after which the two threads sit in the same row at column  $2\lambda$  and read identical cells forever (for  $a \geq 2\lambda$ ,  $\pi(a - \lambda) + e = \pi(a)$ ); the remaining cost is a binomial shift inequality  $f(m - j, k - 1 - j) \geq ((k - 1 - j)/m)^j f(m, k - 1)$  for  $\Pr(\text{Bin}(m, \frac{1}{2}) \leq k - 1) = f(m, k - 1)$ . The loss is *linear* in the lag, not Gaussian; the  $\sqrt{k}$  threshold seen in the simulations is an artefact of the unconditioned regime (both threads at speed 2 with a diffusive offset), whereas conditioned on failure a thread’s runs are geometric at scale  $m/k$  and coalescence is cheap. The mechanism uses no long zero-run, so it survives every global event so far considered  $(\mathcal{A}_{L,s}, \mathcal{B}_{L,s,\rho})$ . (An exact column-by-column Markov computation of  $\Pr(E_0 \cap E_\Delta)$  confirms the rate: 2.0–2.3 nats per unit lag at  $C = 8$ , independent of  $k \in \{100, 400\}$ , and  $\approx \ln C$  across  $C \in [3, 24]$ ; the best shift  $\Delta = h/2$  is worth 45% of an independent thread. For a *random*  $\pi$  at  $C = 8$  every shift has  $\Pr(E_0 \cap E_\Delta)/\Pr(E_0)^2 = e^{140 \pm 10}$  at  $k = 100$  — the run-concentration effect that the global events exist to remove, so some global event is indispensable

in any thread argument.) Consequently (heuristically beyond two threads, since a follower rides whichever earlier thread has the smallest lag and the lags to nearest earlier shifts sum to  $\leq (k/2)(1 + \log_2 \ell)$ )  $\ell$  threads on the tilted grid yield an exponent at most  $Ck \ln 2 + O(k \ln C \log \ell)$ , while the family  $\mathcal{F}(\sqrt{k}, \sqrt{k}, 1/16)$  needs  $\geq (1/32)k \ln k$  per pattern: **the thread framework with polylogarithmically many threads at  $m = Ck$  cannot prove  $n = O(k^2)$** , whatever lag lemma one proves — the same  $k \log \log k$  barrier He and Kwan hit. Details: `experiments/w14-entropy/proof.md`, `experiments/w15-symmetry/proof.md`, `experiments/w18-lag/proof.md`.

*What remains, precisely.* The  $\log \log k$  is **not** removed. With  $\ell \approx \ln k$  threads the chains must be shorter than  $k/(\ell \ln k)$ , so He–Kwan’s structured decomposition must tolerate  $\gtrsim \ln^3 k$  exceptional shifts, whose count is  $e^{\Theta(k \log \log k)}$  — tight for unions of  $r \approx \ln^2 k$ – $\ln^3 k$  increasing runs on value intervals. After Theorem 5 and the symmetry extension, the class still unhandled is exactly the patterns with long shift chains for most shifts in every dihedral image, of which the perturbed tilted grids above are an  $e^{\Theta(k \ln k)}$ -size example; by (iii) these *are* hard for the threads themselves, not only for the chain bound — yet by Theorem 10 each of them individually has threshold  $\leq 0.4k^2$ , so the difficulty is purely one of aggregation. For these, amplification fails (chains of length  $\approx 1.3k/\sqrt{r}$  are too long) and the union bound fails (it would need  $\approx \log r$  independent threads). A per-pattern bound  $\Pr(\pi \not\subseteq \sigma) \leq e^{-k \log r}$  at  $n = Ck^2$  for exactly these  $\pi$  would complete the proof of  $n = O(k^2)$ . We also showed that per-pattern chain accounting alone cannot give it: conditioned on a thread’s failure its zero-runs are geometric at scale  $m/k$ , so a heavy chain costs only  $e^{-\Theta(k)}$  and every later thread rides for free — some global event is unavoidable, and its price is the union bound over the objects it controls. Since (iii) shows that the lag mechanism cannot be the way out, a proof of  $n = O(k^2)$  for the perturbed tilted grids must leave the thread framework: either  $\Theta(\log k)$  threads at  $m = Ck$  with genuinely independent exposures (e.g. threads in the eight dihedral images, or threads restricted to disjoint column blocks), or the point-process methods of the previous section (canonical copies, comparison with the identity), which handle all patterns at once.

## 8 Verification and reproducibility

- **Theorem A** is a short, self-contained proof (§2). The constant  $\lambda_A$  is the root of an explicit one-variable convex minimization; it was computed independently twice (SciPy and mpmath at 30 digits) and the finite- $k$  inequality (A) was evaluated directly at  $k = 10^5$ .
- **Theorems 2–4** ship with witnesses that any reader can re-check: a pure-Python brute force for Theorem 2 (`experiments/witnesses/check_witness.py`: all  $\binom{23}{7} = 245,157$ ,  $\binom{30}{8} = 5,852,925$  and  $\binom{24}{7} = 346,104$  subsets are rank-standardised and counted; self-tests on `sp(3)`–`sp(6)` witnesses and on damaged witnesses; SHA-256 of `witnesses.txt`: `42b52a62...e7be3e4`), a C depth-first checker with different standardisation and enumeration code, and the Lean kernel-checked checker of (i) below, the SAT scripts and UNSAT logs for Theorem 3, the C checker for Theorem 4. All code is in `experiments/`; the witnesses, two independent checkers and their SHA-256 hashes are in `experiments/witnesses/`.
- **Lean 4.** A Lean 4 project (`formal-verification/lean/`, Lean and Mathlib v4.33.1) builds completely (`lake build`, 8715 jobs) and prints an axiom audit. It contains:
  - (i) *Certificates.* Definitions of order-isomorphism, pattern containment and `IsSuperpattern k` ; a DFS checker with the kernel-checked soundness theorem `checker_sound : checker k = true → IsSuperpattern k` (axioms `propext`, `Classical.choice`, `Quot.sound` only); and the theorems `IsSuperpattern 7` (length 23) and `IsSuperpattern 8` for the witnesses of lengths 30, 31 and 32, discharged by `native_decide` (about 5 s for  $k = 7$  and 67 s CPU for  $k = 8$ ,  $n = 30$ ). `native_decide` adds one per-

theorem axiom recording trust in the Lean compiler; the  $k = 3, 4$  examples are proved by the kernel’s `decide` alone, while kernel `decide` for  $k \geq 5$  did not finish (the kernel evaluates array updates as list operations).

- (ii) *Theorem A.* `theoremA` — for `.Nodup`,  $k = 2m + 1$  and  $0 < x < 1$ , `patCount k (x^(n+1)) ^ 1 (x/(1-x))^(k+1) (1-x^k)^m` — is **fully proved**: `lake build` reports that `theoremA`, `lemma1_concrete` and `patCount_le_sum_W` depend only on `propext`, `Classical.choice` and `Quot.sound`. The formalization comprises the finite form of Lemma 2 (`pair_sum_le`), the tilt factorization (`tilt_bound`), the gap bijection (`compSum_gaps`, `gaps_mem_box`, `gaps_injOn`), the abstract fibre-counting form of Lemma 1, the CKS injectivity in counting form (`card_patterns_over_le`) and the extension count (`fiber_card_ge`).
- (iii) *Constant.* `constant_certificate`, i.e.  $1.0003 \cdot 7.37/e^2 - \log 7.37 + 1 + \frac{1}{2} \log(1 - e^{-7.37}) < 0$ , sorry-free, from Mathlib’s rational bounds on  $e$  and  $\log 2$ , the Taylor bound for  $\log(1 - 63/800)$ , and `exp_bound` for  $e^{0.37}$  (hence  $e^{7.37} \leq 1588$ ). The true value is  $\approx -1.25 \times 10^{-5}$ , so the bounds had to be tight. Reproduce with `cd formal-verification/lean && lake exe cache get && lake build`; `formal-verification/lean/LOG.md` records versions, timings and the `#print axioms` output.

## 9 Novelty audit

Before calling a result new we searched under the neighbouring terminology (superpatterns; universal and  $k$ -universal permutations; universal/complete sequences for patterns; dense packing of patterns; zigzag and circular constructions; OEIS A342474), read the papers citing [2] and [3] up to August 2026, and compared every witness with the known ones under the dihedral symmetries (reverse, complement, inverse): the length-23 and length-30 witnesses are not tie-broken words over  $[k + 1]$  in any symmetric image (minimal monotone-row alphabets 9–10 and 13), so they are not disguised Engen–Vatter or Arnarson-type constructions, whereas Arnarson’s 17 *is* a tie-broken three-block word (§4). The unpublished 15/32 upper-bound claim of Hunter is recorded but not used. Details: `memory/priority-check.md`. What has not yet been done is to ask specialists in permutation patterns about prior art or a definitional mismatch; until then the results should be read as a preprint-stage claim.

## 10 Discovery method and AI disclosure

*Declaration of AI-assisted research.* This work was carried out on 2026-08-29 with Claude Code (model Claude Fable 5, `claude-fable-5`) as the discovery engine: a coordinating session read the literature, ranked attack lines, and spawned parallel workstreams (`experiments/w1-w18`), each of which wrote code, ran searches, and drafted proofs. The human author set the goals, chose which directions to pursue, and questioned or rejected claims; the coordinating session independently re-derived every theorem produced by a workstream, recomputed every numerical certificate with separately written code, and brute-forced combinatorial lemmas per class before anything entered this paper — one lemma (an early draft of the value-slot refinement in §2) was false and was caught in this way, and the entropy hypothesis and the lag lemma of §7 were shown to be false rather than assumed. The simulated-annealing witnesses were found with random seeds recorded in `experiments/w1-search/logs/`; SAT results use PySAT/CaDiCaL with the encodings in `experiments/w4-alphabet-rosary/` and `experiments/w6-exact-sp7/`. All workstream briefs, logs, dead ends, code and outputs are archived in the repository; the LLM is not an author, and the

human author takes full responsibility for every claim, citation and program. Every bibliographic entry was checked against arXiv/DOI metadata.

## 11 Open problems

1. Is  $\text{sp}(7) = 23$ ? A proof would need exhaustive search over  $S_{22}$  modulo the dihedral symmetries — the  $k = 6$ ,  $n = 16$  case was done by Pantone;  $n = 22$  is currently out of reach: a SAT/CEGAR formulation (order-matrix variables plus per-pattern prefix chains) re-proves  $\text{sp}(4) = 9$  instantly — the 14 layered and co-layered 4-patterns alone form a minimal unsatisfiable core at length 8 — and  $\text{sp}(5) = 13$  in about 20 s, but the cost grows by more than two orders of magnitude from  $(5, 12)$  to  $(6, 16)$ , which did not finish. Note also that the 226 layered and near-monotone 7-patterns *are* simultaneously realizable at length 22, so any unsatisfiable core there must be larger.
2. Certify  $\lambda = 1.000437$  (two-parity rate) rigorously.
3. Improve Theorem C: width-dependent  $\varepsilon$  already gives 1.00501; both parities (as in Theorem B), keeping  $\prod f(b_i)$  on the bad event, or an exact large deviation for the Poisson-binomial slot count in place of Chernoff should push further towards the heuristic ceiling 1.0117; and formalize Theorem C in Lean.
4.  $f(6; 7)$ ;  $r(6)$ ,  $r(7)$ ; Gupta’s conjecture for odd  $n \geq 13$ .

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